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Course Number	MTE604
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Semester/Year	3
Section Number	04
Group Number	

Project Report

Assignment Title	Project Report
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Abstract

This project investigated the modeling and control of a single-axis solar tracking system using Gazebo simulation and a Python-based feedback controller. The objective was to design a solar panel mechanism capable of following a prescribed sun trajectory over a 25-second interval while maintaining stable and accurate motion. The system was modeled as a rotational mass-damper system and controlled using a PD-equivalent state-feedback controller with gains $K_p = 12, \quad K_d = 15$.

The tracking performance of the system was evaluated under five scenarios: a nominal baseline case, wind disturbance, sensor inaccuracy, actuator saturation, and gearbox implementation. Performance was assessed by comparing the reference and actual panel angles and by calculating the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for each case. The results showed that the controller achieved the best tracking performance in the nominal scenario, while the introduction of disturbance, measurement error, and actuator limits increased tracking error to varying degrees. Even so, the system remained stable in all tested scenarios.

Overall, the project demonstrated that a properly designed feedback controller can provide effective solar tracking under ideal conditions and maintain acceptable performance under more realistic operating effects. The study also highlighted the importance of considering disturbances, imperfect sensing, and actuator limitations when evaluating electromechanical control systems for renewable energy applications.

Introduction

Solar tracking systems are used to improve the energy capture of photovoltaic panels by continuously adjusting panel orientation so that the panel remains aligned with the sun. Compared with fixed solar panels, tracking systems can increase the amount of solar radiation received throughout the day, which makes them important in renewable energy applications. In practice, however, accurate solar tracking depends on both the mechanical design of the tracker and the performance of its control system. A useful tracking system must not only follow the desired sun trajectory, but also remain stable and accurate when realistic effects such as disturbance, sensor error, and actuator limitations are present.

The objective of this project was to model and simulate a single-axis solar tracking system in Gazebo and evaluate its tracking performance under several operating conditions. The system was implemented as a rotating solar panel mounted on a fixed support and controlled using a Python-based feedback controller. The panel dynamics were modeled as a second-order rotational mass-damper system, and a PD-equivalent state-feedback controller was used to track the prescribed sun reference trajectory over a 25-second simulation period. The controller gains selected for the final implementation were $K_p = 12, \quad K_d = 15$.

To assess both nominal performance and robustness, five scenarios were studied. Scenario 1 considered the ideal baseline case, while the remaining scenarios introduced wind disturbance, sensor inaccuracy, actuator saturation, and gearbox effects. The performance of the solar tracker was evaluated

using plots of reference and actual panel motion, together with tracking error metrics including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Through these scenarios, the project investigated how realistic physical limitations and uncertainties affect the ability of the controller to maintain accurate solar tracking.

Solar Tracking System

The solar tracking system was developed in Gazebo as a simplified single-axis solar panel mechanism that rotates to follow the sun's path. The purpose of the design was to simulate a practical renewable energy tracking system and then evaluate how well a feedback controller could make the panel follow a desired angular reference over time. According to the project requirements, the system consisted of a mechanical model built in Gazebo and a Python-based controller used to command the panel motion.

Mechanical System Design

The mechanical model was created as a single-axis tracking structure with two main bodies: a fixed support pole and a solar panel link. In the Gazebo model, the base pole served as the stationary support, while the solar panel was mounted above it and allowed to rotate about one axis. The panel motion was achieved using a revolute joint connecting the base pole to the panel. This configuration represents a single-axis solar tracker, where the panel rotates to follow the apparent motion of the sun across the sky.

The Gazebo world also included the required simulation environment, such as the ground plane, lighting, and physics plugins. The system used a high-resolution physics time step so that the motion of the panel and the controller response could be simulated smoothly. The revolute joint was constrained to rotate within the limits given in the project specification, and damping and friction were included to make the motion more realistic.

Mathematical Modeling

To design the controller, the solar panel was modeled as a second-order rotational mechanical system. The project specification states that the panel should be treated as a rotational mass-damper system of the form

$$I\ddot{\theta} + b\dot{\theta} = \tau$$

Where I is the panel moment of inertia, b is the viscous damping coefficient, and τ is the applied joint torque. This model captures the main dynamics of the tracker because the panel rotates about a single axis and its motion is affected by inertia and damping.

From the project description, the solar panel inertia values were defined in the SDF model, and the joint damping was also specified in Gazebo. These physical properties were used as the basis for the controller design. The rotational model is appropriate because the system only needs to track the desired panel angle as the sun reference changes over the 25-second simulation interval.

Reference Trajectory

The desired sun motion was defined using the reference trajectory required by the project:

$$\theta_{ref}(t) = -\cos(\frac{\pi t}{T})$$

Where T = 25s is the total simulation time. This function generates a smooth angular motion from −1 rad to +1 rad, representing the panel tracking the sun across the sky. During simulation, the controller continuously compared the panel angle with this reference and adjusted the commanded motion accordingly.

Controller Design

The solar tracking system was controlled using a state-feedback controller equivalent to a PD controller, as required by the project. The controller gains were obtained theoretically by modeling the panel as a second-order rotational mass-damper system,

$$I\ddot{\theta} + b\dot{\theta} = \tau$$

and then matching the closed-loop equation to the standard critically damped second-order form. By choosing a damping ratio of $\zeta = 1$ and a suitable natural frequency, the proportional and derivative gains could be determined from the system parameters. In the final implementation, the selected gains were:

$$K_p = 12, \quad K_d = 15$$

These gains were used to reduce tracking error, improve damping, and produce a stable response as the panel followed the changing sun reference. The controller was implemented in Gazebo through the joint position controller plugin, while a Python script generated the reference trajectory and monitored the panel motion during each scenario.

Results

Scenario #1:

Scenario 1 represents the nominal operating condition of the solar tracking system, where the panel tracks the desired sun trajectory without any external disturbances or sensor inaccuracies. This scenario serves as the baseline case for evaluating the controller performance under ideal conditions.

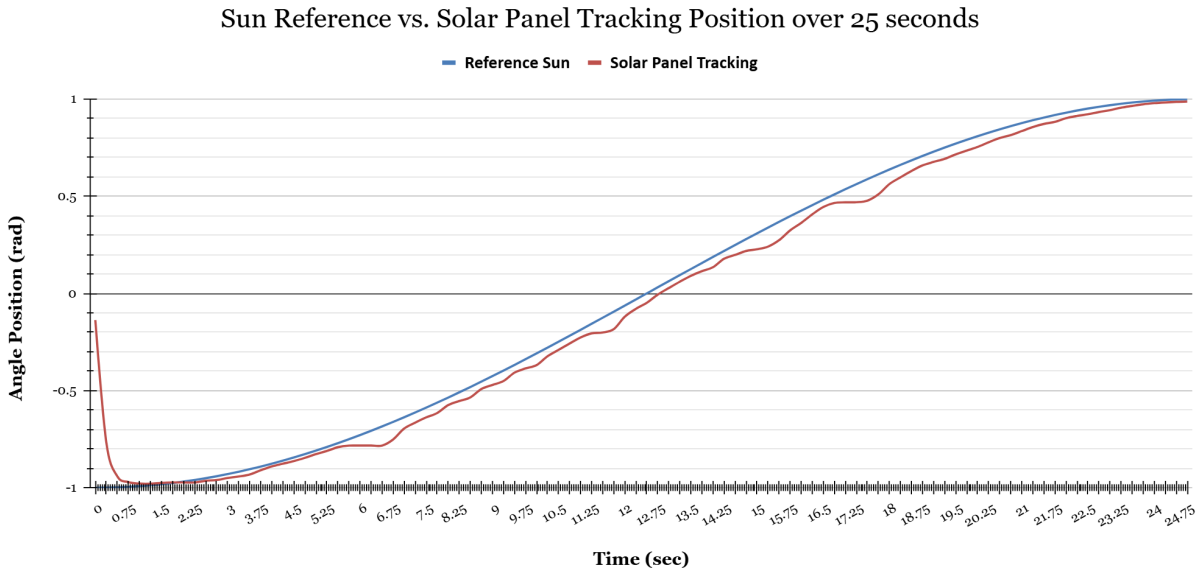


Figure 1. Scenario 1 Tracking Graph

Figure 1 shows the reference sun trajectory and the actual solar panel position over the 25-second simulation. Overall, the solar panel tracks the reference trajectory closely, showing that the controller is able to regulate the panel motion effectively. After a short initial transient, the actual position follows the desired path with relatively small deviation for most of the simulation.

We can calculate the Mean Absolute Error (MAE) using our tracking error data from Table 1 in the Appendix, along with the following formula:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|e_i\right|$$

After calculating, our Mean Absolute Error is: 0.0534215 rad.

Similarly, we can calculate the Root Mean Squared Error (RMSE) using our tracking data error from Table 1, this time using the formula:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^{\,2}}$$

After calculating with the full data set, our Root Mean Squared Error is: 0.10192 rad.

Mean Absolute Error	Root Mean Squared Error
0.0534215	0.1019198506

Table A. Calculated MAE and RMSE for Table 1 Data

The results can be justified by considering the second-order dynamics of the solar tracking mechanism. At the start of the simulation, a relatively large tracking error is observed because the panel must move from its initial condition to match the prescribed reference trajectory. This produces a short transient response before the system settles into regular tracking. After this initial adjustment, the error decreases significantly, indicating that the controller is effective in regulating the panel motion. The remaining small deviation between the actual and reference trajectories can be attributed to the mechanical inertia and damping of the system, which limit how quickly the panel can respond to continuous changes in the desired position. Overall, the results indicate that the controller gains were appropriately selected, since the system remains stable and achieves accurate tracking under ideal operating conditions.

Scenario #2:

Scenario 2 evaluates the performance of the solar tracking system when an external wind disturbance is applied to the panel. In this case, the controller must track the desired sun trajectory while also rejecting the effects of disturbance torque acting on the tracking joint. The wind disturbance is modeled as a combination of a smooth oscillating component, high-frequency random turbulence, and occasional gust spikes, making the operating condition more realistic and more challenging than the nominal case. This scenario is used to determine how well the controller maintains tracking accuracy and stability when the system is subjected to environmental disturbances.

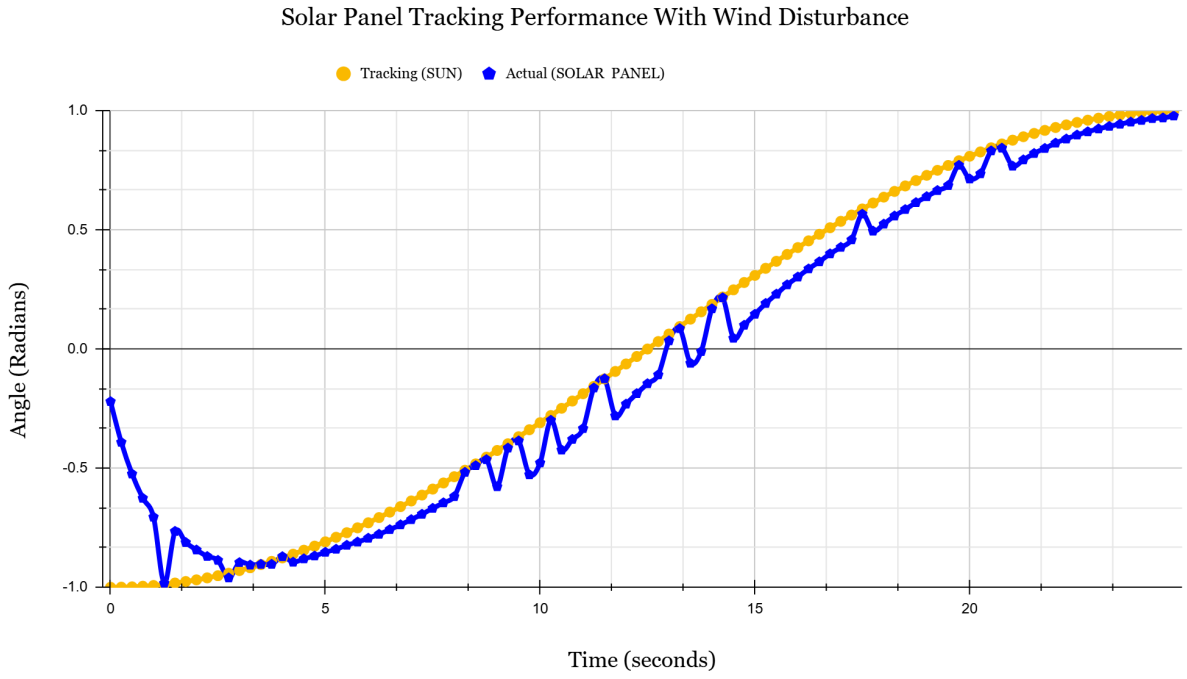


Figure 2. Solar Panel Tracking with Wind Disturbance

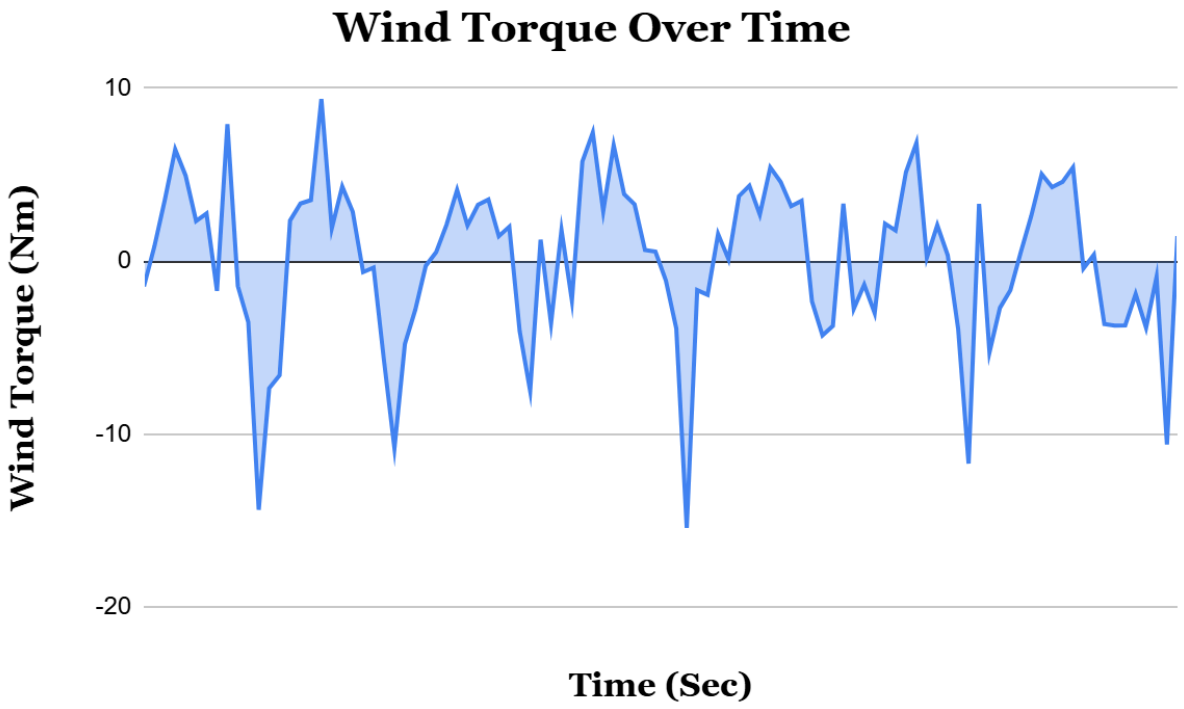


Figure 3. Graph of Wind Torque Over Time

Figure 2 shows that the solar panel is still able to follow the overall sun reference trajectory when wind disturbance is introduced, but the tracking response is noticeably less smooth than in Scenario 1. The actual panel position exhibits several oscillations and sudden dips throughout the simulation, especially during the middle portion of the motion, indicating that the disturbance affects the panel's ability to remain closely aligned with the desired path. Even with these deviations, the overall trend of the response still matches the reference trajectory, and the panel eventually approaches the final target angle. Figure 3 further shows that the applied wind torque varies continuously with both positive and negative fluctuations, along with several larger spikes that represent stronger gust disturbances.

This behavior can be explained by the effect of wind torque acting as an external disturbance on the tracking joint. As the disturbance changes over time, especially during sudden gusts, the controller must not only track the reference signal but also counteract the additional disturbance torque. Because the mechanical system has inertia and damping, the panel cannot respond instantaneously to these rapid changes, which leads to temporary lag and larger tracking deviations than in the nominal case. The repeated fluctuations in the actual panel angle are consistent with the varying wind torque shown in Figure 3. However, the system remains stable throughout the simulation, which indicates that the controller is still effective in rejecting much of the disturbance while maintaining acceptable overall tracking performance.

The Mean Absolute Error (MSE) and Root Mean Squared Error (RMSE) are calculated with the same respective formulas used in Scenario #1. This time the tracking error data from Table 2 in the appendix is used.

After calculating, the Mean Absolute Error for the error data in Table 2 is: 0.1009114 rad, and the Root Mean Squared Error for the data is: 0.15284 rad.

MAE	RMSE
0.1009114	0.152840599

Table B. Calculated MAE and RMSE for Table 2 Data

Compared with Scenario 1, both the MAE and RMSE increase in Scenario 2, which confirms that the wind disturbance reduces the tracking accuracy of the system.

The increase in tracking error in Scenario 2 is justified by the presence of the external wind disturbance torque acting on the panel. Unlike Scenario 1, the controller must now both follow the desired sun trajectory and continuously reject the effects of the changing disturbance. Since the wind torque fluctuates over time and includes stronger gust-like spikes, the actual panel motion shows temporary deviations and oscillations from the reference trajectory. In addition, the inertia and damping of the mechanical system prevent the panel from responding instantaneously to these rapid changes. As a result, the MAE and RMSE are larger than in the nominal case, although the controller still maintains stable overall tracking.

Scenario #3:

Scenario 3 evaluates the performance of the solar tracking system when the position measurement is affected by sensor inaccuracy. In this case, the controller no longer relies on perfect joint angle data, and the measured signal is corrupted by Gaussian noise, quantization, and slow drift. These imperfections simulate the behavior of a realistic sensor, where measurements may fluctuate, lose resolution, or gradually shift over time. The purpose of this scenario is to determine how well the controller can maintain tracking accuracy and stability when the feedback signal itself is imperfect.

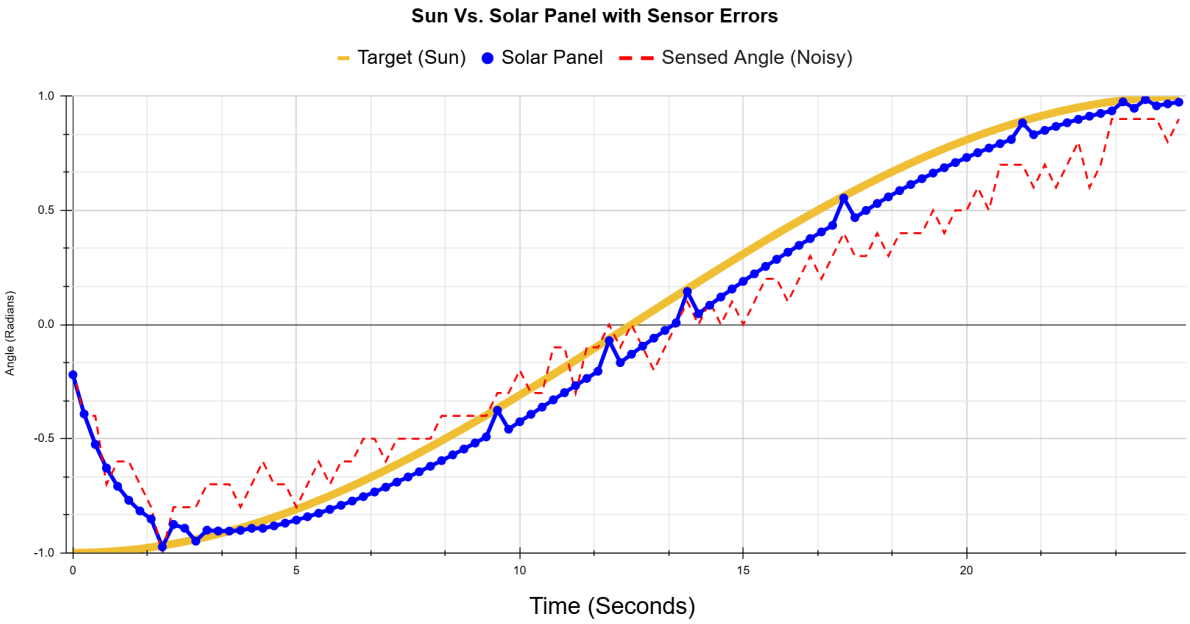


Figure 4. Plot Sun vs. Solar Panel with Sensor Errors

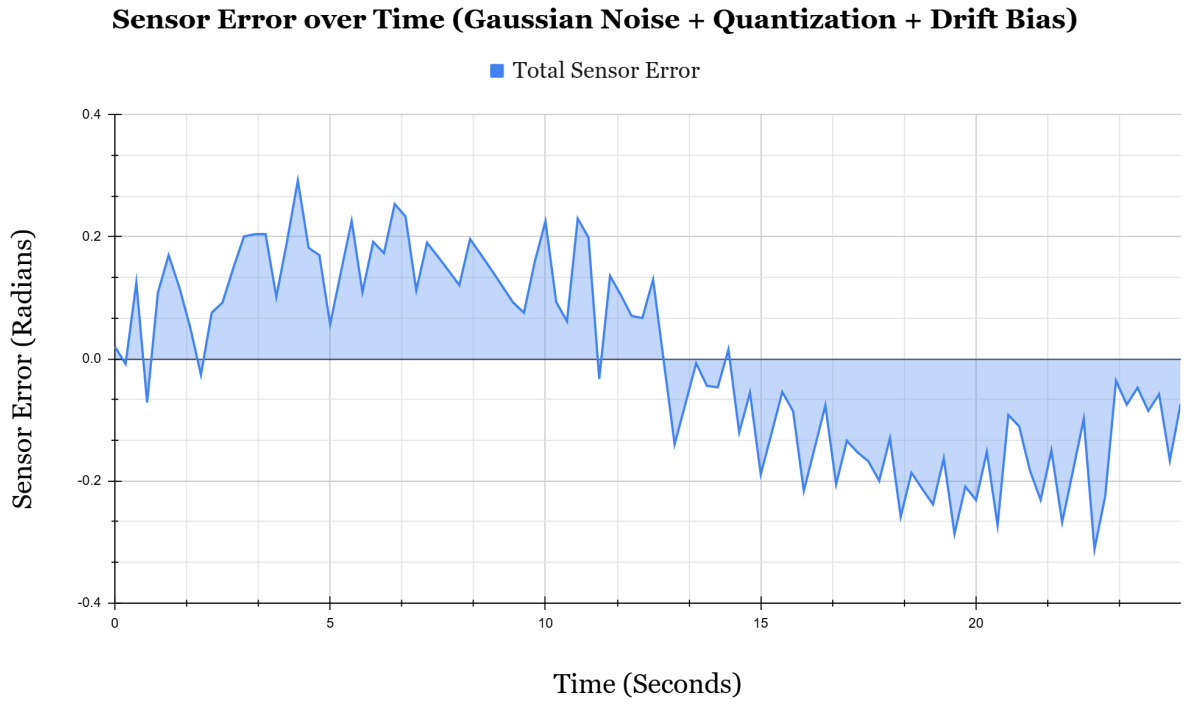


Figure 5. Plot of Sensor Error over Time

Figure 4 shows that the solar panel is still able to follow the overall sun reference trajectory even when sensor inaccuracies are introduced, but the sensed angle is noticeably noisier than the true panel

motion. The measured signal fluctuates around the actual response and does not remain perfectly aligned with it throughout the simulation. Despite this imperfect feedback, the true panel angle still follows the general trend of the target trajectory and reaches the desired final position with only moderate deviation. Figure 5 shows that the total sensor error varies over time and includes both positive and negative offsets, indicating that the measurement corruption is not constant and changes throughout the simulation.

This behavior is caused by the controller relying on a corrupted feedback signal rather than the true panel angle. Because the measurement includes Gaussian noise, quantization, and drift, the sensed angle contains random fluctuations, reduced resolution, and a slowly varying bias. These effects make the controller respond to inaccurate position information, which introduces additional tracking deviation compared to the ideal case. The variations seen in the sensor error plot explain why the sensed response is less smooth and less accurate than the true panel motion. However, the system remains stable and the actual panel trajectory still follows the desired path reasonably well, which shows that the controller can tolerate a certain amount of sensor imperfection while maintaining acceptable tracking performance.

The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are calculated with the same formulas used in Scenarios #1 and #2. This time the tracking error data from Table 3 in the Appendix is used.

After performing calculations, the Mean Absolute Error for the error data in Table 3 is: 0.1871 rad, and the Root Mean Squared Error for the data is: 0.22944 rad.

MAE	RMSE
0.1870604	0.2294405727

Table C. Calculated MAE and RMSE for Table 3 Data

The higher tracking error in Scenario 3 is justified by the fact that the controller is no longer using perfect position feedback. Instead, the measured angle is corrupted by Gaussian noise, quantization, and drift, which causes the controller to respond to inaccurate information rather than the true panel position. As a result, the sensed signal fluctuates and contains bias, leading to larger deviations between the reference trajectory and the actual panel motion. Quantization reduces the resolution of the measurement, noise introduces random variation, and drift causes the measured value to gradually shift over time. Together, these effects degrade tracking accuracy and increase both the MAE and RMSE compared with the ideal baseline case. However, the system still remains stable and follows the overall reference trend, showing that the controller can tolerate sensor imperfections even though its performance is reduced.

Scenario #4:

Scenario 4 evaluates the performance of the solar tracking system when actuator limitations are introduced. In this case, the motor can no longer provide unlimited control effort, since the applied torque is capped at a maximum value and the joint speed is also restricted to a low rotational rate. These limits simulate a more realistic actuator with saturation and gearbox constraints, where the panel may be unable to respond quickly enough when large control action is required. The purpose of this scenario is to examine how torque and speed saturation affect the system’s tracking accuracy, transient response, and overall stability.

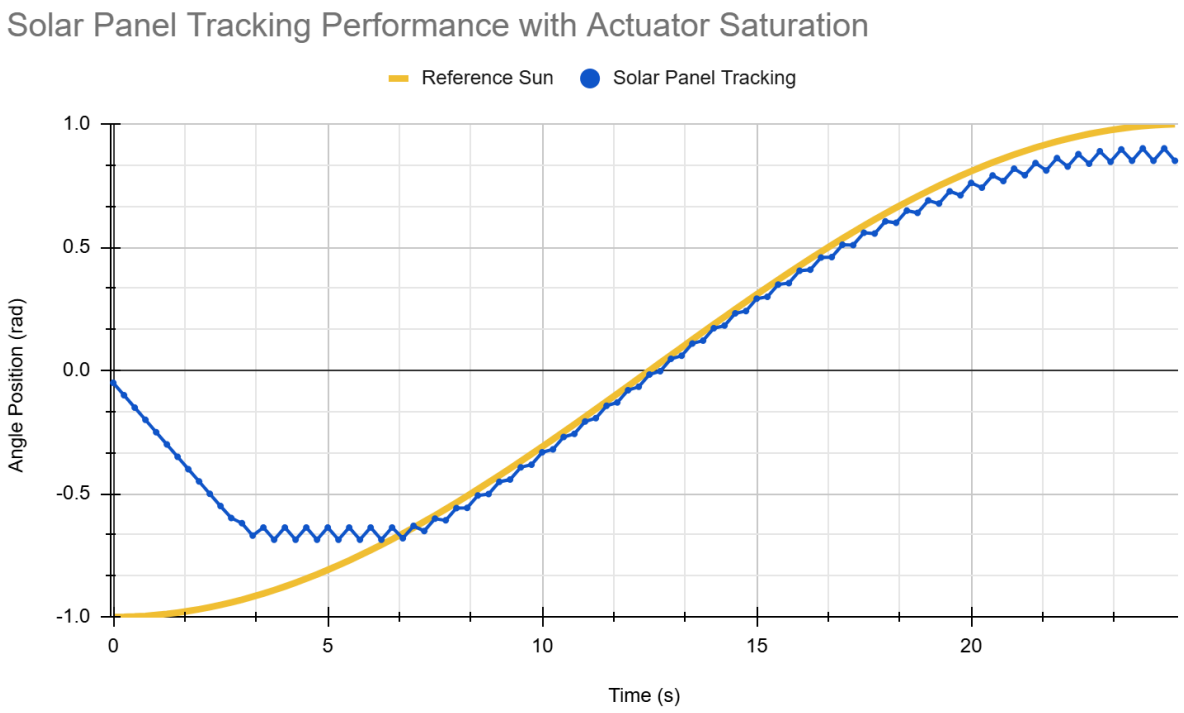


Figure 6. Plot of Solar Panel Tracking Performance with Actuator Saturation

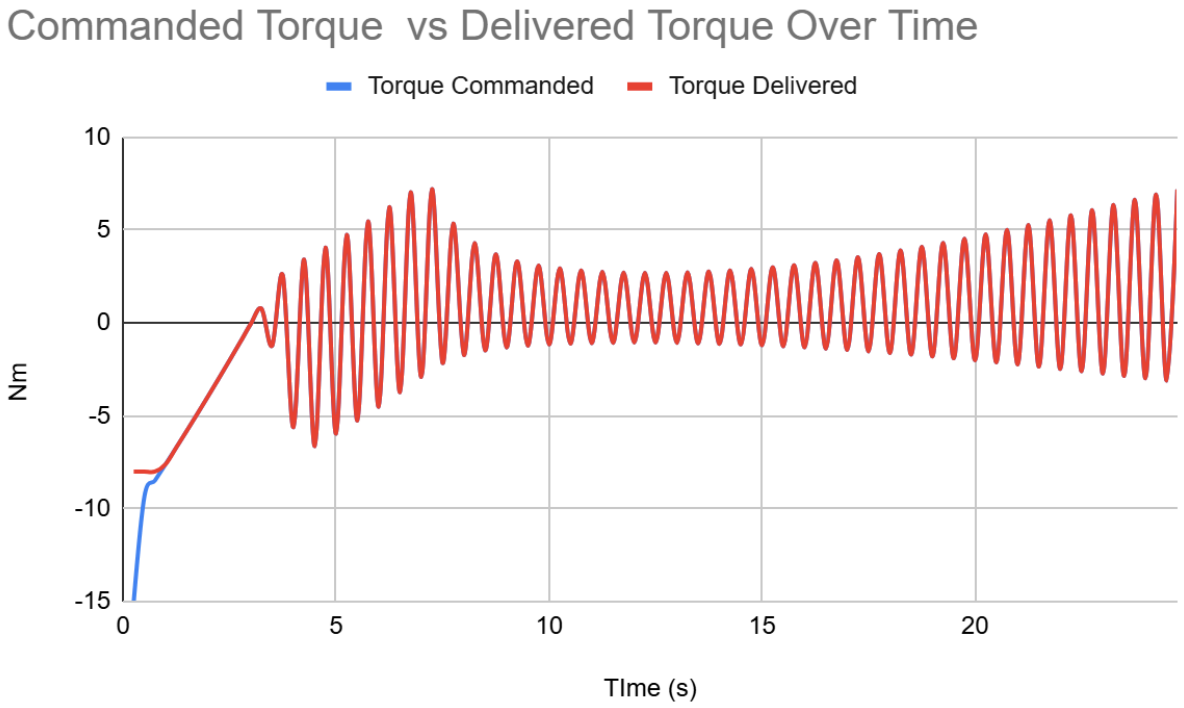


Figure 7. Plot of Commanded Torque vs Delivered Torque

Figure 6 shows that the solar panel still follows the overall reference trajectory when actuator saturation is introduced, but the response is slower and less accurate than in the earlier scenarios. The actual panel position lags behind the reference for most of the simulation, and small oscillations are visible as the panel approaches the desired motion. Although the system remains stable, the actuator limits prevent the panel from tracking the reference as closely as before.

This behavior is caused by the torque and speed restrictions on the actuator. Since the motor cannot provide unlimited effort and the joint cannot rotate faster than the imposed speed limit, the panel cannot respond immediately to the changing reference. This leads to increased lag and reduced tracking accuracy, even though the system still maintains stable operation.

Figure 7 shows the commanded torque and the delivered torque under actuator saturation. At several points in the simulation, the controller demands a larger torque than the actuator is allowed to provide, so the delivered torque is limited instead of fully matching the command. This confirms that the actuator is operating under saturation constraints, which explains why the panel cannot always respond fast enough to track the reference trajectory exactly. The difference between the commanded and delivered torque contributes directly to the lag and reduced tracking accuracy observed in Figure 6.

The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are calculated with the same formulas used in Scenarios #1, #2, and #3. This time the tracking error data from Table 4 in the Appendix is used.

After performing calculations, the Mean Absolute Error and Root Mean Squared Error for the error data in Table 4 is:

MAE	RMSE
0.1395446	0.2518351679

Table D. Calculated MAE and RMSE for Table 4 Data.

The larger error in Scenario 4 is justified by the actuator saturation limits. Since the motor torque and joint speed are restricted, the panel cannot respond as quickly as needed to follow the reference trajectory exactly. This causes increased lag and small oscillations in the actual response, which lead to higher MAE and RMSE values than in the ideal case. Even with these limitations, the system remains stable and continues to follow the overall reference trend.

Scenario #5

Scenario 5 evaluates the performance of the solar tracking system when a mechanical gearbox is introduced into the actuator model. In this case, a weak motor is paired with a 10:1 gear ratio so that the motor's torque is amplified before driving the panel. This setup better represents a practical solar tracking mechanism, since real solar tracking systems often rely on gear reduction to allow a small motor to move a larger mechanical load. By increasing the available output torque, the gearbox helps the motor rotate the panel more effectively, even though the motor itself has limited torque capacity. The purpose of this scenario is to examine how the gearbox affects the panel's tracking accuracy, response speed, and overall dynamic behavior, and to determine whether the added torque amplification is sufficient to maintain stable and acceptable tracking performance.

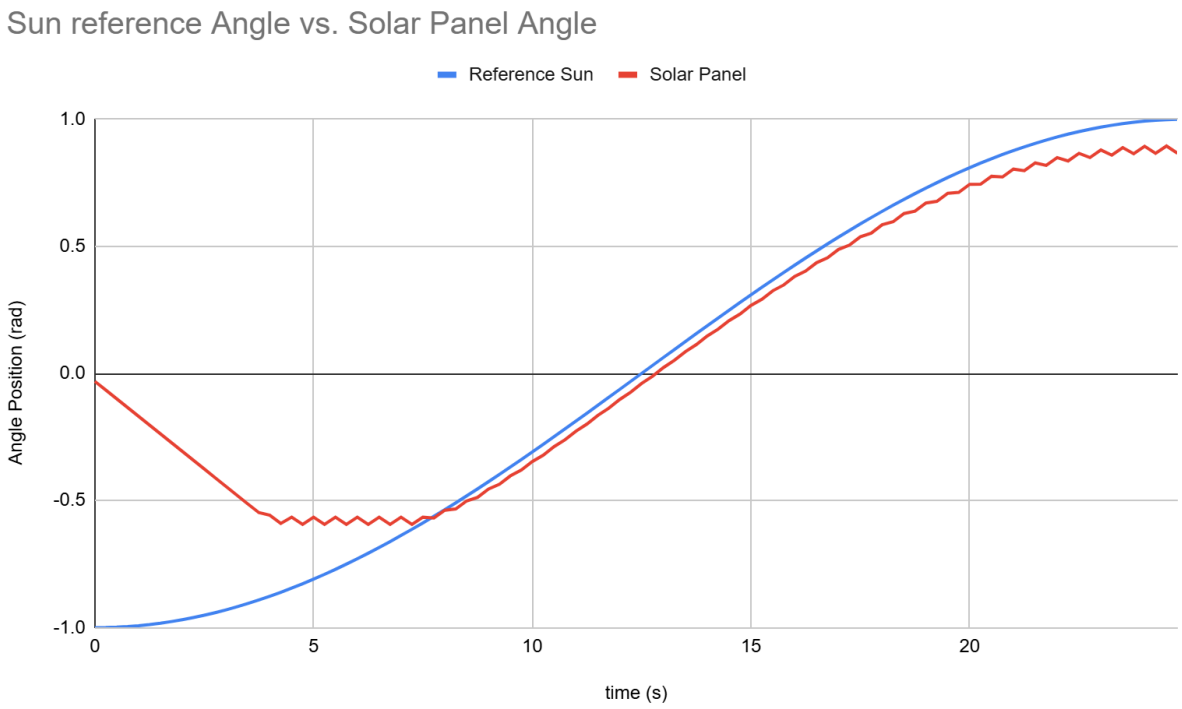


Figure 8. Sun reference vs. actual Solar Panel angle over time

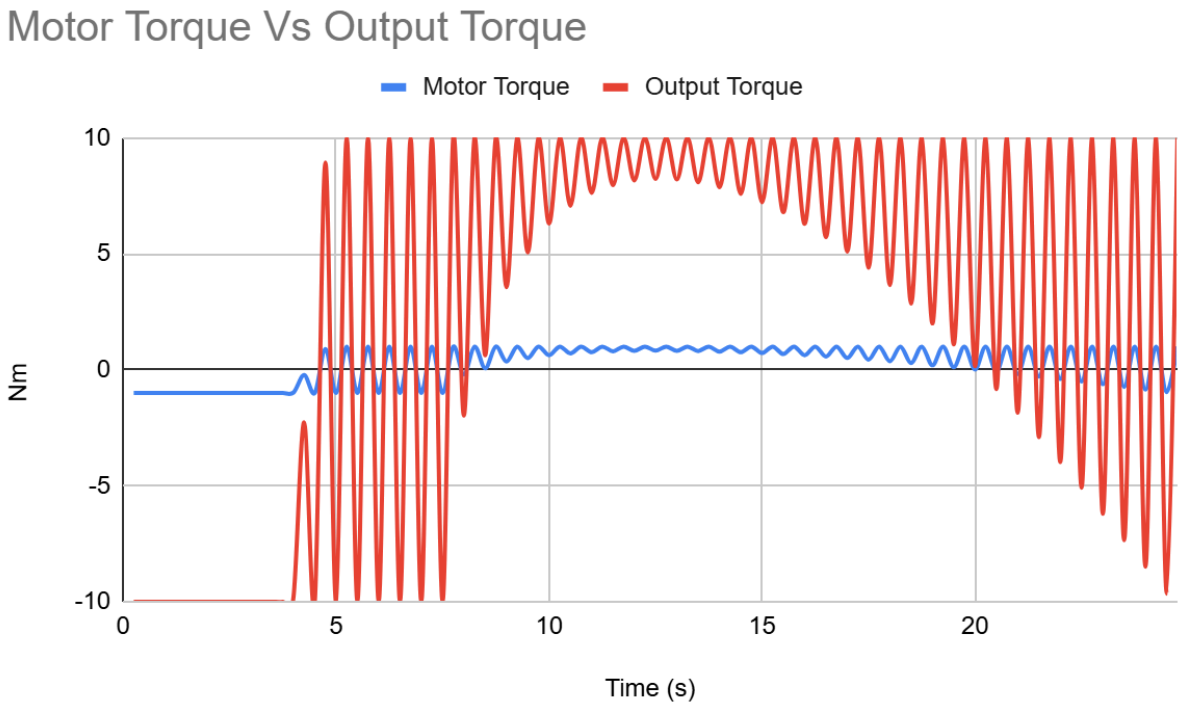


Figure 9. Plot of Motor Torque vs Output Torque over Time

Figure 8 shows that the solar panel still follows the overall reference trajectory when the gearbox is introduced, although the actual response lags slightly behind the desired motion. The panel angle remains close to the sun reference for most of the simulation, with only small oscillations visible near the end.

This behavior is due to the gearbox amplifying the weak motor's torque before it drives the panel. As a result, the system is able to maintain stable and acceptable tracking, although some lag and tracking error still remain.

Figure 9 shows the motor torque and gearbox output torque over time. As expected, the output torque has a much larger magnitude than the motor torque due to the 10:1 gearbox ratio. This demonstrates how the gearbox amplifies the weak motor torque before it is applied to the solar panel, allowing the system to generate enough rotational effort to track the reference motion.

After performing calculations, the Mean Absolute Error and Root Mean Squared Error for the error data in Table 5 is:

MAE	RMSE
0.1733383	0.2944010051

Table E. Calculated MAE and RMSE for Table 5 Data.

The tracking error in Scenario 5 is caused by the limited performance of the weak motor, even with the added gearbox. Although the 10:1 gear ratio increases the output torque and helps the panel follow the reference motion, the response still shows some lag because the system cannot adjust instantly to the changing trajectory. This leads to higher MAE and RMSE values than the ideal case. Even so, the gearbox allows the system to remain stable and achieve acceptable overall tracking performance.

Conclusion

In this project, a single-axis solar tracking system was successfully modeled in Gazebo and controlled using a PD-based feedback controller to follow a prescribed sun trajectory. The results showed that the controller performed best in the nominal case, where the panel tracked the reference closely with relatively low error. This confirmed that the selected controller gains were suitable for the baseline system and that the modeled tracker could achieve stable and accurate motion under ideal conditions.

When non-ideal effects were introduced, the tracking performance degraded to different degrees. Wind disturbance caused oscillations and temporary deviations from the reference, sensor inaccuracy reduced feedback quality and increased tracking error, actuator saturation introduced lag by limiting control effort and speed, and the gearbox scenario showed that torque amplification could still allow acceptable tracking with a weak motor, although some error remained. Across all scenarios, the system remained stable, but the MAE and RMSE values increased compared with the baseline case, showing that real-world disturbances and limitations have a significant effect on performance.

Overall, the project demonstrated that effective solar tracking depends on both proper controller design and realistic consideration of system limitations. The simulations showed that the solar tracker can follow the sun trajectory successfully, but also highlighted the tradeoff between ideal tracking accuracy and practical operating conditions. This project provided useful insight into the modeling, control, and performance evaluation of electromechanical tracking systems used in renewable energy applications.

Appendix

time_s	sun_reference_rad	panel_actual_rad	tracking_error_rad
0	-1	-0.13717	-0.86283
0.25	-0.99951	-0.76836	-0.23115
0.5	-0.99803	-0.94186	-0.05617
0.75	-0.99556	-0.97113	-0.02443
1	-0.99211	-0.97871	-0.0134
1.25	-0.98769	-0.97927	-0.00842
1.5	-0.98229	-0.97529	-0.007
1.75	-0.97592	-0.97285	-0.00307
2	-0.96858	-0.97285	0.00427
2.25	-0.96029	-0.97285	0.01256
2.5	-0.95106	-0.96452	0.01346
2.75	-0.94088	-0.96037	0.01949
3	-0.92978	-0.94996	0.02019
3.25	-0.91775	-0.9424	0.02465
3.5	-0.90483	-0.93176	0.02693
3.75	-0.89101	-0.91005	0.01904
4	-0.87631	-0.89027	0.01397
4.25	-0.86074	-0.87664	0.01589
4.5	-0.84433	-0.86316	0.01883
4.75	-0.82708	-0.84622	0.01914
5	-0.80902	-0.827	0.01798
5.25	-0.79016	-0.81082	0.02066
5.5	-0.77051	-0.79139	0.02088
5.75	-0.75011	-0.78321	0.0331
6	-0.72897	-0.78294	0.05397
6.25	-0.70711	-0.78294	0.07583
6.5	-0.68455	-0.78294	0.09839
6.75	-0.66131	-0.7517	0.09039
7	-0.63742	-0.69541	0.05798
7.25	-0.61291	-0.66522	0.05231
7.5	-0.58779	-0.6381	0.05031
7.75	-0.56208	-0.61555	0.05346
8	-0.53583	-0.57468	0.03885
8.25	-0.50904	-0.55392	0.04488
8.5	-0.48175	-0.53561	0.05386
8.75	-0.45399	-0.49231	0.03832

9	-0.42578	-0.4712	0.04542
9.25	-0.39715	-0.45116	0.05401
9.5	-0.36812	-0.40798	0.03986
9.75	-0.33874	-0.38625	0.04751
10	-0.30902	-0.3694	0.06039
10.25	-0.27899	-0.32339	0.0444
10.5	-0.24869	-0.29055	0.04186
10.75	-0.21814	-0.25795	0.03981
11	-0.18738	-0.22663	0.03925
11.25	-0.15643	-0.20562	0.04918
11.5	-0.12533	-0.20141	0.07607
11.75	-0.09411	-0.18399	0.08989
12	-0.06279	-0.1203	0.05751
12.25	-0.03141	-0.08005	0.04864
12.5	0	-0.04837	0.04837
12.75	0.03141	-0.00586	0.03727
13	0.06279	0.02764	0.03515
13.25	0.09411	0.06029	0.03382
13.5	0.12533	0.09055	0.03479
13.75	0.15643	0.11455	0.04189
14	0.18738	0.13463	0.05275
14.25	0.21814	0.17823	0.03991
14.5	0.24869	0.19792	0.05077
14.75	0.27899	0.21815	0.06084
15	0.30902	0.22704	0.08198
15.25	0.33874	0.24089	0.09784
15.5	0.36812	0.27398	0.09414
15.75	0.39715	0.32422	0.07293
16	0.42578	0.36267	0.06311
16.25	0.45399	0.40664	0.04735
16.5	0.48175	0.44436	0.03739
16.75	0.50904	0.46503	0.04402
17	0.53583	0.46854	0.06729
17.25	0.56208	0.46909	0.09299
17.5	0.58779	0.47713	0.11065
17.75	0.61291	0.50966	0.10325
18	0.63742	0.5628	0.07463
18.25	0.66131	0.59713	0.06418
18.5	0.68455	0.6299	0.05464

18.75	0.70711	0.65857	0.04854
19	0.72897	0.67644	0.05253
19.25	0.75011	0.6923	0.05781
19.5	0.77051	0.71522	0.05529
19.75	0.79016	0.73494	0.05521
20	0.80902	0.75378	0.05523
20.25	0.82708	0.77766	0.04942
20.5	0.84433	0.79934	0.04498
20.75	0.86074	0.81474	0.046
21	0.87631	0.83518	0.04113
21.25	0.89101	0.85604	0.03497
21.5	0.90483	0.87215	0.03268
21.75	0.91775	0.8826	0.03516
22	0.92978	0.90145	0.02832
22.25	0.94088	0.91313	0.02775
22.5	0.95106	0.92185	0.02921
22.75	0.96029	0.93314	0.02716
23	0.96858	0.9429	0.02568
23.25	0.97592	0.95556	0.02036
23.5	0.98229	0.96455	0.01774
23.75	0.98769	0.97319	0.0145
24	0.99211	0.97886	0.01325
24.25	0.99556	0.98269	0.01287
24.5	0.99803	0.98562	0.01241
24.75	0.99951	0.98737	0.01214

Table 1. Full Scenario #1 Tracking Data

time_s	sun_reference_rad	panel_actual_rad	tracking_error_rad	wind_torque
0	-1	-0.22045	-0.77955	-1.46519
0.25	-0.99951	-0.39143	-0.60807	0.83323
0.5	-0.99803	-0.52419	-0.47383	3.52552
0.75	-0.99556	-0.62589	-0.36967	6.46163
1	-0.99211	-0.70522	-0.2869	4.94446
1.25	-0.98769	-0.98253	-0.00515	2.324
1.5	-0.98229	-0.7651	-0.21719	2.75399
1.75	-0.97592	-0.81155	-0.16437	-1.72276
2	-0.96858	-0.843	-0.12558	7.92834

2.25	-0.96029	-0.87035	-0.08995	-1.44632
2.5	-0.95106	-0.88689	-0.06417	-3.53295
2.75	-0.94088	-0.96006	0.01918	-14.38632
3	-0.92978	-0.89595	-0.03383	-7.35547
3.25	-0.91775	-0.90622	-0.01153	-6.60577
3.5	-0.90483	-0.9031	-0.00172	2.34916
3.75	-0.89101	-0.90352	0.01251	3.33379
4	-0.87631	-0.87101	-0.00529	3.53434
4.25	-0.86074	-0.89425	0.03351	9.38282
4.5	-0.84433	-0.88047	0.03615	1.89215
4.75	-0.82708	-0.86843	0.04135	4.31754
5	-0.80902	-0.85296	0.04395	2.86923
5.25	-0.79016	-0.83999	0.04983	-0.63588
5.5	-0.77051	-0.82316	0.05264	-0.37224
5.75	-0.75011	-0.80987	0.05976	-5.75209
6	-0.72897	-0.79358	0.06462	-10.82251
6.25	-0.70711	-0.77713	0.07003	-4.79517
6.5	-0.68455	-0.75762	0.07308	-2.8078
6.75	-0.66131	-0.73771	0.07639	-0.27125
7	-0.63742	-0.71554	0.07812	0.52828
7.25	-0.61291	-0.69307	0.08016	2.11923
7.5	-0.58779	-0.66854	0.08075	4.11577
7.75	-0.56208	-0.64501	0.08293	2.06627
8	-0.53583	-0.61923	0.0834	3.25908
8.25	-0.50904	-0.51717	0.00813	3.56053
8.5	-0.48175	-0.48935	0.0076	1.44104
8.75	-0.45399	-0.46382	0.00983	1.98873
9	-0.42578	-0.57738	0.1516	-4.06663
9.25	-0.39715	-0.41538	0.01824	-7.50288
9.5	-0.36812	-0.38545	0.01733	1.24155
9.75	-0.33874	-0.52722	0.18848	-3.60741
10	-0.30902	-0.47752	0.16851	1.7971
10.25	-0.27899	-0.29901	0.02002	-2.26691
10.5	-0.24869	-0.42353	0.17484	5.76309
10.75	-0.21814	-0.3788	0.16065	7.44007

11	-0.18738	-0.33354	0.14616	2.89276
11.25	-0.15643	-0.16297	0.00653	6.70661
11.5	-0.12533	-0.12469	-0.00064	3.87524
11.75	-0.09411	-0.28036	0.18625	3.28706
12	-0.06279	-0.23088	0.16809	0.643
12.25	-0.03141	-0.18709	0.15568	0.54367
12.5	0	-0.14532	0.14532	-1.12488
12.75	0.03141	-0.1079	0.13931	-3.89726
13	0.06279	0.03441	0.02838	-15.43326
13.25	0.09411	0.08575	0.00836	-1.67652
13.5	0.12533	-0.05944	0.18478	-1.95339
13.75	0.15643	-0.01051	0.16694	1.57155
14	0.18738	0.16907	0.01831	0.11396
14.25	0.21814	0.21546	0.00268	3.75826
14.5	0.24869	0.04574	0.20295	4.36507
14.75	0.27899	0.09971	0.17928	2.69467
15	0.30902	0.14608	0.16293	5.4236
15.25	0.33874	0.1913	0.14744	4.576
15.5	0.36812	0.23041	0.13772	3.1758
15.75	0.39715	0.26973	0.12742	3.48439
16	0.42578	0.30255	0.12323	-2.34159
16.25	0.45399	0.33657	0.11742	-4.29866
16.5	0.48175	0.36625	0.11551	-3.74257
16.75	0.50904	0.39986	0.10918	3.32778
17	0.53583	0.42687	0.10895	-2.75595
17.25	0.56208	0.45845	0.10363	-1.36106
17.5	0.58779	0.56636	0.02143	-3.00962
17.75	0.61291	0.49525	0.11765	2.1643
18	0.63742	0.52426	0.11316	1.77107
18.25	0.66131	0.5584	0.10291	5.15248
18.5	0.68455	0.58561	0.09894	6.8382
18.75	0.70711	0.61488	0.09223	0.20052
19	0.72897	0.63944	0.08953	2.07724
19.25	0.75011	0.66554	0.08457	0.34194
19.5	0.77051	0.68651	0.084	-3.92763

19.75	0.79016	0.77126	0.01889	-11.70901
20	0.80902	0.71391	0.0951	3.31333
20.25	0.82708	0.73594	0.09114	-5.27814
20.5	0.84433	0.83171	0.01262	-2.70009
20.75	0.86074	0.84324	0.0175	-1.68789
21	0.87631	0.768	0.10831	0.5234
21.25	0.89101	0.79414	0.09687	2.61741
21.5	0.90483	0.82104	0.08379	5.03132
21.75	0.91775	0.84224	0.07551	4.28232
22	0.92978	0.86441	0.06536	4.58973
22.25	0.94088	0.88188	0.059	5.4267
22.5	0.95106	0.89842	0.05263	-0.42406
22.75	0.96029	0.91207	0.04823	0.36734
23	0.96858	0.92412	0.04446	-3.64393
23.25	0.97592	0.93451	0.04141	-3.73121
23.5	0.98229	0.944	0.03829	-3.7169
23.75	0.98769	0.95279	0.0349	-1.90203
24	0.99211	0.95994	0.03217	-3.84761
24.25	0.99556	0.96753	0.02803	-0.92469
24.5	0.99803	0.97018	0.02785	-10.60637
24.75	0.99951	0.97832	0.02118	1.45455

Table 2. Full Scenario #2 Data

time_s	sun_reference_rad	panel_true_rad	panel_measured_rad	tracking_error_rad	gaussian_noise_rad	drift_rad	total_sensor_error_rad
0	-1	-0.22	-0.2	-0.8	0.04073	0	0.02
0.25	-0.99951	-0.39149	-0.4	-0.59951	0.01996	0.01199	-0.00851
0.5	-0.99803	-0.52493	-0.4	-0.59803	0.09056	0.02394	0.12493
0.75	-0.99556	-0.62847	-0.7	-0.29556	-0.08701	0.03581	-0.07153
1	-0.99211	-0.70847	-0.6	-0.39211	0.03399	0.04754	0.10847
1.25	-0.98769	-0.7699	-0.6	-0.38769	0.14071	0.0591	0.1699
1.5	-0.98229	-0.81662	-0.7	-0.28229	0.04869	0.07045	0.11662
1.75	-0.97592	-0.85167	-0.8	-0.17592	-0.03348	0.08155	0.05167
2	-0.96858	-0.97381	-1	0.03142	-0.08535	0.09236	-0.02619

2.25	-0.96029	-0.87557	-0.8	-0.16029	-0.00214	0.10283	0.07557
2.5	-0.95106	-0.89217	-0.8	-0.15106	-0.07035	0.11293	0.09217
2.75	-0.94088	-0.94846	-0.8	-0.14088	0.01891	0.12262	0.14846
3	-0.92978	-0.90045	-0.7	-0.22978	0.03764	0.13188	0.20045
3.25	-0.91775	-0.90425	-0.7	-0.21775	0.05323	0.14066	0.20425
3.5	-0.90483	-0.90438	-0.7	-0.20483	0.03836	0.14893	0.20438
3.75	-0.89101	-0.90144	-0.8	-0.09101	-0.02709	0.15667	0.10144
4	-0.87631	-0.89264	-0.7	-0.17631	-0.00944	0.16384	0.19264
4.25	-0.86074	-0.89248	-0.6	-0.26074	0.09637	0.17042	0.29248
4.5	-0.84433	-0.88189	-0.7	-0.14433	-0.03247	0.17639	0.18189
4.75	-0.82708	-0.86983	-0.7	-0.12708	-0.01508	0.18173	0.16983
5	-0.80902	-0.85645	-0.8	-0.00902	-0.09385	0.18641	0.05645
5.25	-0.79016	-0.84187	-0.7	-0.09016	-0.06287	0.19042	0.14187
5.5	-0.77051	-0.82617	-0.6	-0.17051	0.00599	0.19374	0.22617
5.75	-0.75011	-0.80944	-0.7	-0.05011	-0.08045	0.19637	0.10944
6	-0.72897	-0.79173	-0.6	-0.12897	0.0064	0.19829	0.19173
6.25	-0.70711	-0.77312	-0.6	-0.10711	-0.06526	0.1995	0.17312
6.5	-0.68455	-0.75363	-0.5	-0.18455	0.0576	0.19999	0.25363
6.75	-0.66131	-0.73332	-0.5	-0.16131	0.03272	0.19976	0.23332
7	-0.63742	-0.71222	-0.6	-0.03742	-0.04795	0.19881	0.11222
7.25	-0.61291	-0.69037	-0.5	-0.11291	-0.01984	0.19714	0.19037
7.5	-0.58779	-0.6678	-0.5	-0.08779	0.00716	0.19477	0.1678
7.75	-0.56208	-0.64455	-0.5	-0.06208	-0.04806	0.19169	0.14455
8	-0.53583	-0.62063	-0.5	-0.03583	-0.08246	0.18793	0.12063
8.25	-0.50904	-0.59608	-0.4	-0.10904	0.0076	0.18349	0.19608
8.5	-0.48175	-0.57093	-0.4	-0.08175	0.00346	0.17839	0.17093
8.75	-0.45399	-0.5452	-0.4	-0.05399	-0.03432	0.17264	0.1452
9	-0.42578	-0.51893	-0.4	-0.02578	-0.07691	0.16628	0.11893
9.25	-0.39715	-0.49214	-0.4	0.00285	-0.08029	0.15931	0.09214
9.5	-0.36812	-0.37552	-0.3	-0.06812	-0.08205	0.15178	0.07552
9.75	-0.33874	-0.45839	-0.3	-0.03874	0.03356	0.14369	0.15839
10	-0.30902	-0.42553	-0.2	-0.10902	0.11039	0.13509	0.22553
10.25	-0.27899	-0.39329	-0.3	0.02101	-0.01037	0.12601	0.09329
10.5	-0.24869	-0.36148	-0.3	0.05131	-0.05959	0.11647	0.06148
10.75	-0.21814	-0.32994	-0.1	-0.11814	0.09342	0.10651	0.22994

11	-0.18738	-0.29858	-0.1	-0.08738	0.07678	0.09616	0.19858
11.25	-0.15643	-0.26731	-0.3	0.14357	-0.07018	0.08548	-0.03269
11.5	-0.12533	-0.23607	-0.1	-0.02533	0.01906	0.07448	0.13607
11.75	-0.09411	-0.20484	-0.1	0.00589	0.07133	0.06322	0.10484
12	-0.06279	-0.07036	0	-0.06279	0.03551	0.05172	0.07036
12.25	-0.03141	-0.16669	-0.1	0.06859	0.04848	0.04005	0.06669
12.5	0	-0.13002	0	0	0.06332	0.02822	0.13002
12.75	0.03141	-0.0945	-0.1	0.13141	-0.06994	0.0163	-0.0055
13	0.06279	-0.0599	-0.2	0.26279	-0.09783	0.00432	-0.1401
13.25	0.09411	-0.02602	-0.1	0.19411	-0.06494	-0.00768	-0.07398
13.5	0.12533	0.00728	0	0.12533	0.03919	-0.01965	-0.00728
13.75	0.15643	0.14423	0.1	0.05643	0.0335	-0.03155	-0.04423
14	0.18738	0.0469	0	0.18738	-0.03267	-0.04334	-0.0469
14.25	0.21814	0.08458	0.1	0.11814	0.08807	-0.05496	0.01542
14.5	0.24869	0.12068	0	0.24869	-0.06096	-0.0664	-0.12068
14.75	0.27899	0.15551	0.1	0.17899	0.04261	-0.07759	-0.05551
15	0.30902	0.18928	0	0.30902	-0.07313	-0.0885	-0.18928
15.25	0.33874	0.22216	0.1	0.23874	0.01519	-0.0991	-0.12216
15.5	0.36812	0.25427	0.2	0.16812	0.04737	-0.10934	-0.05427
15.75	0.39715	0.28571	0.2	0.19715	-0.01186	-0.11918	-0.08571
16	0.42578	0.31652	0.1	0.32578	-0.06152	-0.1286	-0.21652
16.25	0.45399	0.34676	0.2	0.25399	0.02628	-0.13755	-0.14676
16.5	0.48175	0.37646	0.3	0.18175	0.06037	-0.14601	-0.07646
16.75	0.50904	0.40563	0.2	0.30904	-0.01595	-0.15394	-0.20563
17	0.53583	0.43427	0.3	0.23583	0.06014	-0.16132	-0.13427
17.25	0.56208	0.55343	0.4	0.16208	0.01653	-0.16812	-0.15343
17.5	0.58779	0.46805	0.3	0.28779	-0.00814	-0.17432	-0.16805
17.75	0.61291	0.49992	0.3	0.31291	-0.01816	-0.17988	-0.19992
18	0.63742	0.53017	0.4	0.23742	0.07902	-0.1848	-0.13017
18.25	0.66131	0.55902	0.3	0.36131	-0.08058	-0.18905	-0.25902
18.5	0.68455	0.58664	0.4	0.28455	0.04749	-0.19263	-0.18664
18.75	0.70711	0.61314	0.4	0.30711	0.00144	-0.19551	-0.21314
19	0.72897	0.63862	0.4	0.32897	-0.07579	-0.19768	-0.23862
19.25	0.75011	0.66315	0.5	0.25011	0.04469	-0.19915	-0.16315
19.5	0.77051	0.68677	0.4	0.37051	-0.13574	-0.1999	-0.28677

19.75	0.79016	0.70951	0.5	0.29016	-0.00408	-0.19992	-0.20951
20	0.80902	0.7314	0.5	0.30902	-0.07318	-0.19923	-0.2314
20.25	0.82708	0.75245	0.6	0.22708	0.04326	-0.19783	-0.15245
20.5	0.84433	0.77267	0.5	0.34433	-0.06099	-0.19571	-0.27267
20.75	0.86074	0.79204	0.7	0.16074	0.0836	-0.19288	-0.09204
21	0.87631	0.81058	0.7	0.17631	0.05131	-0.18936	-0.11058
21.25	0.89101	0.88274	0.7	0.19101	0.046	-0.18516	-0.18274
21.5	0.90483	0.83131	0.6	0.30483	-0.06432	-0.1803	-0.23131
21.75	0.91775	0.85033	0.7	0.21775	0.06568	-0.17478	-0.15033
22	0.92978	0.86781	0.6	0.32978	-0.07096	-0.16864	-0.26781
22.25	0.94088	0.88389	0.7	0.24088	0.02266	-0.16189	-0.18389
22.5	0.95106	0.89866	0.8	0.15106	0.02498	-0.15455	-0.09866
22.75	0.96029	0.91222	0.6	0.36029	-0.16051	-0.14666	-0.31222
23	0.96858	0.92462	0.7	0.26858	-0.07153	-0.13825	-0.22462
23.25	0.97592	0.93591	0.9	0.07592	0.07719	-0.12933	-0.03591
23.5	0.98229	0.97523	0.9	0.08229	0.00376	-0.11995	-0.07523
23.75	0.98769	0.9473	0.9	0.08769	0.01944	-0.11014	-0.0473
24	0.99211	0.98541	0.9	0.09211	0.03978	-0.09993	-0.08541
24.25	0.99556	0.95792	0.9	0.09556	0.01187	-0.08936	-0.05792
24.5	0.99803	0.96674	0.8	0.19803	-0.08408	-0.07847	-0.16674
24.75	0.99951	0.97395	0.9	0.09951	0.00957	-0.0673	-0.07395

Table 3. Full Scenario #3 Data

time_s	sun_reference_rad	panel_actual_rad	tracking_error_rad	torque_command_Nm	torque_delivered_Nm
0	-1	-0.05	-0.95	-15	-8
0.25	-0.99951	-0.1	-0.89951	-9.34787	-8
0.5	-0.99803	-0.15	-0.84803	-8.48103	-8
0.75	-0.99556	-0.2	-0.79556	-7.59961	-7.59961
1	-0.99211	-0.25	-0.74211	-6.70372	-6.70372
1.25	-0.98769	-0.3	-0.68769	-5.79353	-5.79353
1.5	-0.98229	-0.35	-0.63229	-4.86918	-4.86918
1.75	-0.97592	-0.4	-0.57592	-3.93085	-3.93085
2	-0.96858	-0.45	-0.51858	-2.97872	-2.97872

2.25	-0.96029	-0.5	-0.46029	-2.01299	-2.01299
2.5	-0.95106	-0.55	-0.40106	-1.03387	-1.03387
2.75	-0.94088	-0.59818	-0.34271	-0.0416	-0.0416
3	-0.92978	-0.61956	-0.31022	0.76102	0.76102
3.25	-0.91775	-0.66956	-0.2482	-1.2225	-1.2225
3.5	-0.90483	-0.63659	-0.26824	2.55507	2.55507
3.75	-0.89101	-0.68659	-0.20442	-5.61232	-5.61232
4	-0.87631	-0.63659	-0.23972	3.40713	3.40713
4.25	-0.86074	-0.68659	-0.17416	-6.6271	-6.6271
4.5	-0.84433	-0.63659	-0.20774	4.04989	4.04989
4.75	-0.82708	-0.68659	-0.14049	-5.96221	-5.96221
5	-0.80902	-0.63659	-0.17243	4.73626	4.73626
5.25	-0.79016	-0.68659	-0.10357	-5.25505	-5.25505
5.5	-0.77051	-0.63659	-0.13393	5.46352	5.46352
5.75	-0.75011	-0.68659	-0.06352	-4.5084	-4.5084
6	-0.72897	-0.63659	-0.09238	6.22881	6.22881
6.25	-0.70711	-0.68659	-0.02052	-3.72522	-3.72522
6.5	-0.68455	-0.63659	-0.04796	7.0291	7.0291
6.75	-0.66131	-0.68061	0.01929	-2.9086	-2.9086
7	-0.63742	-0.63061	-0.00682	7.19734	7.19734
7.25	-0.61291	-0.65096	0.03806	-2.15148	-2.15148
7.5	-0.58779	-0.60096	0.01318	5.34205	5.34205
7.75	-0.56208	-0.60791	0.04583	-1.72238	-1.72238
8	-0.53583	-0.55791	0.02209	4.29486	4.29486
8.25	-0.50904	-0.55701	0.04797	-1.47099	-1.47099
8.5	-0.48175	-0.50701	0.02525	3.68464	3.68464
8.75	-0.45399	-0.5013	0.04731	-1.31755	-1.31755
9	-0.42578	-0.4513	0.02553	3.31442	3.31442
9.25	-0.39715	-0.44255	0.0454	-1.21976	-1.21976
9.5	-0.36812	-0.39255	0.02442	3.07977	3.07977
9.75	-0.33874	-0.38177	0.04304	-1.15523	-1.15523
10	-0.30902	-0.33177	0.02276	2.92535	2.92535
10.25	-0.27899	-0.31965	0.04066	-1.11209	-1.11209
10.5	-0.24869	-0.26965	0.02096	2.82186	2.82186
10.75	-0.21814	-0.25665	0.03851	-1.08409	-1.08409

11	-0.18738	-0.20665	0.01927	2.75386	2.75386
11.25	-0.15643	-0.19315	0.03672	-1.06791	-1.06791
11.5	-0.12533	-0.14315	0.01782	2.71315	2.71315
11.75	-0.09411	-0.12946	0.03535	-1.06181	-1.06181
12	-0.06279	-0.07946	0.01667	2.69543	2.69543
12.25	-0.03141	-0.06586	0.03445	-1.06485	-1.06485
12.5	0	-0.01586	0.01586	2.69838	2.69838
12.75	0.03141	-0.00263	0.03404	-1.07651	-1.07651
13	0.06279	0.04737	0.01542	2.72073	2.72073
13.25	0.09411	0.05999	0.03412	-1.09645	-1.09645
13.5	0.12533	0.10999	0.01535	2.76172	2.76172
13.75	0.15643	0.12172	0.03471	-1.12447	-1.12447
14	0.18738	0.17172	0.01566	2.82082	2.82082
14.25	0.21814	0.18233	0.03581	-1.16037	-1.16037
14.5	0.24869	0.23233	0.01636	2.89762	2.89762
14.75	0.27899	0.24159	0.03741	-1.20397	-1.20397
15	0.30902	0.29159	0.01743	2.99171	2.99171
15.25	0.33874	0.29924	0.0395	-1.25508	-1.25508
15.5	0.36812	0.34924	0.01889	3.10267	3.10267
15.75	0.39715	0.35507	0.04208	-1.31348	-1.31348
16	0.42578	0.40507	0.02071	3.23002	3.23002
16.25	0.45399	0.40885	0.04514	-1.37894	-1.37894
16.5	0.48175	0.45885	0.0229	3.37326	3.37326
16.75	0.50904	0.46038	0.04867	-1.4512	-1.4512
17	0.53583	0.51038	0.02545	3.5318	3.5318
17.25	0.56208	0.50944	0.05264	-1.52997	-1.52997
17.5	0.58779	0.55944	0.02835	3.70503	3.70503
17.75	0.61291	0.55585	0.05706	-1.61493	-1.61493
18	0.63742	0.60585	0.03158	3.89225	3.89225
18.25	0.66131	0.59942	0.06189	-1.70575	-1.70575
18.5	0.68455	0.64942	0.03513	4.09272	4.09272
18.75	0.70711	0.63998	0.06713	-1.80208	-1.80208
19	0.72897	0.68998	0.03899	4.30565	4.30565
19.25	0.75011	0.67737	0.07274	-1.90353	-1.90353
19.5	0.77051	0.72737	0.04315	4.5302	4.5302

19.75	0.79016	0.71144	0.07872	-2.0097	-2.0097
20	0.80902	0.76144	0.04758	4.76549	4.76549
20.25	0.82708	0.74206	0.08502	-2.12018	-2.12018
20.5	0.84433	0.79206	0.05227	5.01058	5.01058
20.75	0.86074	0.76911	0.09164	-2.23452	-2.23452
21	0.87631	0.81911	0.0572	5.26451	5.26451
21.25	0.89101	0.79247	0.09854	-2.35227	-2.35227
21.5	0.90483	0.84247	0.06236	5.52628	5.52628
21.75	0.91775	0.81207	0.10569	-2.47298	-2.47298
22	0.92978	0.86207	0.06771	5.79485	5.79485
22.25	0.94088	0.82781	0.11307	-2.59617	-2.59617
22.5	0.95106	0.87781	0.07325	6.06916	6.06916
22.75	0.96029	0.83964	0.12065	-2.72134	-2.72134
23	0.96858	0.88964	0.07894	6.34813	6.34813
23.25	0.97592	0.84752	0.1284	-2.848	-2.848
23.5	0.98229	0.89752	0.08477	6.63066	6.63066
23.75	0.98769	0.8514	0.13628	-2.97566	-2.97566
24	0.99211	0.9014	0.09071	6.91564	6.91564
24.25	0.99556	0.8514	0.14416	-3.10381	-3.10381
24.5	0.99803	0.9014	0.09662	7.18871	7.18871
24.75	0.99951	0.8514	0.1481	-3.23373	-3.23373

Table 4. Full Scenario #4 Data

time_s	sun_reference_rad	panel_actual_rad	tracking_error_rad	motor_torque_Nm	output_torque_Nm
0	-1	-0.03125	-0.96875	-1	-10
0.25	-0.99951	-0.06543	-0.93408	-1	-10
0.5	-0.99803	-0.09988	-0.89814	-1	-10
0.75	-0.99556	-0.13436	-0.8612	-1	-10
1	-0.99211	-0.16885	-0.82327	-1	-10
1.25	-0.98769	-0.20333	-0.78436	-1	-10
1.5	-0.98229	-0.23781	-0.74448	-1	-10
1.75	-0.97592	-0.27229	-0.70362	-1	-10
2	-0.96858	-0.30678	-0.66181	-1	-10

2.25	-0.96029	-0.34126	-0.61903	-1	-10
2.5	-0.95106	-0.37574	-0.57531	-1	-10
2.75	-0.94088	-0.41023	-0.53065	-1	-10
3	-0.92978	-0.44471	-0.48507	-1	-10
3.25	-0.91775	-0.47919	-0.43856	-1	-10
3.5	-0.90483	-0.51367	-0.39115	-1	-10
3.75	-0.89101	-0.54755	-0.34346	-0.98044	-9.80437
4	-0.87631	-0.55781	-0.3185	-0.22683	-2.26825
4.25	-0.86074	-0.59002	-0.27072	-1	-10
4.5	-0.84433	-0.56511	-0.27922	0.89379	8.93794
4.75	-0.82708	-0.59402	-0.23306	-1	-10
5	-0.80902	-0.56549	-0.24353	1	10
5.25	-0.79016	-0.59406	-0.19609	-1	-10
5.5	-0.77051	-0.56549	-0.20502	1	10
5.75	-0.75011	-0.59406	-0.15605	-1	-10
6	-0.72897	-0.56549	-0.16348	1	10
6.25	-0.70711	-0.59406	-0.11305	-1	-10
6.5	-0.68455	-0.56549	-0.11906	1	10
6.75	-0.66131	-0.59406	-0.06725	-1	-10
7	-0.63742	-0.56549	-0.07193	1	10
7.25	-0.61291	-0.59406	-0.01885	-1	-10
7.5	-0.58779	-0.56549	-0.0223	1	10
7.75	-0.56208	-0.56898	0.00689	-0.19736	-1.97356
8	-0.53583	-0.53805	0.00223	1	10
8.25	-0.50904	-0.5332	0.02416	0.06254	0.62539
8.5	-0.48175	-0.5015	0.01974	1	10
8.75	-0.45399	-0.4874	0.0334	0.35612	3.56123
9	-0.42578	-0.45482	0.02904	1	10
9.25	-0.39715	-0.43595	0.03881	0.5061	5.06104
9.5	-0.36812	-0.40293	0.03481	1	10
9.75	-0.33874	-0.38013	0.04139	0.63076	6.30761
10	-0.30902	-0.34674	0.03772	1	10
10.25	-0.27899	-0.32151	0.04252	0.70715	7.0715
10.5	-0.24869	-0.2879	0.03921	1	10
10.75	-0.21814	-0.26092	0.04277	0.76253	7.62534

11	-0.18738	-0.22714	0.03975	1	10
11.25	-0.15643	-0.19908	0.04264	0.7965	7.96499
11.5	-0.12533	-0.1652	0.03987	1	10
11.75	-0.09411	-0.13651	0.0424	0.81638	8.16379
12	-0.06279	-0.10257	0.03978	1	10
12.25	-0.03141	-0.07365	0.04224	0.82365	8.23648
12.5	0	-0.03969	0.03969	1	10
12.75	0.03141	-0.01086	0.04227	0.82064	8.20644
13	0.06279	0.02309	0.0397	1	10
13.25	0.09411	0.05154	0.04257	0.80831	8.08311
13.5	0.12533	0.08545	0.03988	1	10
13.75	0.15643	0.11324	0.04319	0.78754	7.87544
14	0.18738	0.1471	0.04028	1	10
14.25	0.21814	0.17399	0.04416	0.75886	7.58858
14.5	0.24869	0.20776	0.04093	1	10
14.75	0.27899	0.23351	0.04548	0.72269	7.22692
15	0.30902	0.26717	0.04185	1	10
15.25	0.33874	0.29156	0.04718	0.67938	6.79376
15.5	0.36812	0.32509	0.04303	1	10
15.75	0.39715	0.3479	0.04925	0.62921	6.2921
16	0.42578	0.38129	0.04449	1	10
16.25	0.45399	0.40231	0.05168	0.57247	5.72471
16.5	0.48175	0.43553	0.04622	1	10
16.75	0.50904	0.45456	0.05448	0.50944	5.09435
17	0.53583	0.4876	0.04823	1	10
17.25	0.56208	0.50446	0.05763	0.44038	4.40383
17.5	0.58779	0.53729	0.0505	1	10
17.75	0.61291	0.55179	0.06112	0.36561	3.65609
18	0.63742	0.5844	0.05302	1	10
18.25	0.66131	0.59638	0.06493	0.28542	2.85422
18.5	0.68455	0.62875	0.0558	1	10
18.75	0.70711	0.63804	0.06907	0.20015	2.00146
19	0.72897	0.67016	0.05881	1	10
19.25	0.75011	0.67661	0.0735	0.11012	1.10124
19.5	0.77051	0.70847	0.06205	1	10

19.75	0.79016	0.71195	0.07821	0.01571	0.15714
20	0.80902	0.74352	0.0655	1	10
20.25	0.82708	0.7439	0.08318	-0.08271	-0.82708
20.5	0.84433	0.77518	0.06915	1	10
20.75	0.86074	0.77234	0.0884	-0.18475	-1.84753
21	0.87631	0.80333	0.07298	1	10
21.25	0.89101	0.79717	0.09384	-0.29002	-2.90017
21.5	0.90483	0.82784	0.07699	1	10
21.75	0.91775	0.81828	0.09948	-0.39808	-3.98084
22	0.92978	0.84863	0.08115	1	10
22.25	0.94088	0.83558	0.1053	-0.50853	-5.08527
22.5	0.95106	0.86561	0.08545	1	10
22.75	0.96029	0.84902	0.11127	-0.62091	-6.20911
23	0.96858	0.87872	0.08987	1	10
23.25	0.97592	0.85854	0.11738	-0.73479	-7.3479
23.5	0.98229	0.8879	0.09439	1	10
23.75	0.98769	0.86409	0.12359	-0.84972	-8.49717
24	0.99211	0.89311	0.099	1	10
24.25	0.99556	0.86567	0.12989	-0.96524	-9.65236
24.5	0.99803	0.89435	0.10368	1	10
24.75	0.99951	0.86579	0.13372	-1	-10

Table 5. Full Scenario #5 Data